Chris Foster

chris@fosterelli.co
Victoria, British Columbia
https://fosterelli.co
+1 (250) 572-7938

Professional Experience

President

February 2026 - Present | Remote

Corunio Software Corp.

Founder of a company focused on designing, acquiring, and operating focused software tools with an emphasis on architecture, reliability, and continuity. Our portfolio includes a student management system that has helped 1 in 20 BC students participate in distance education and a health services product that has helped Canadian medical specialists triage more than 40,000 cases.

Founder / CTO

January 2020 - Present | Remote

Path and Focus Technologies Inc.

Co-founder and CTO of a climate startup on a mission to help humanity adapt to our changing environment. We do that by building software technologies to quantify and reduce wildfire risk. My position involves all areas of the company but I primarily focus on leading our engineering team, designing our technical architecture, and iterating on our products with our customers and stakeholders.

Chief Technology Officer

June 2019 - April 2026 | Remote

Two Story Robot Labs Inc.

Technical leadership role as the co-owner and CTO of a boutique agency specializing in web application development, IoT, and machine learning. My position focused on engineering and architecture while also involving management, coaching, business development, hiring, strategic planning, and product responsibilities. I worked with modern technologies including React, GraphQL, Node, Docker, AWS, MongoDB, Postgres, and Kubernetes.

Entrepreneur in Residence

Feb 2024 - Aug 2025 | Remote

Central Interior Business Accelerator

Provided coaching services to startup clients. My role worked closely with other EiRs and the executive director to deliver CIBA’s accelerator program. My focus was on coaching startups with significant technical requirements but my role also included guidance on product validation, early stage customer acquisition, and co-founder relationships.

Entrepreneur in Residence

May 2023 - Feb 2024 | Remote

Kamloops Innovation Centre Society

Provided coaching services to startup clients and facilitated discussions with potential clients. My role worked closely with other EiRs and the executive director to deliver KICS's accelerator program. I primarily worked one-on-one with founders assigned to me but also provided advice to other clients through quarterly review sessions as well as in specific situations with accelerator clients who needed technical leadership guidance.

Software Developer

September 2016 - June 2019 | *Remote*

Two Story Robot Labs Inc.

Full stack developer for multiple SaaS products, and consulting for both government and private organizations. While I primarily focused on engineering and architecture, the consulting positions often required support, research, data science, and mentoring roles. I developed with modern technologies including Javascript, Node, AWS, C++, Elasticsearch, RethinkDB, Angular, RabbitMQ, CouchDB, Docker, React, Apollo, GraphQL, and Redux.

Software Developer

September 2012 - September 2016 | *Kamloops, B.C.*

MemoryLeaf Media Inc.

Full stack web application developer for multiple company SaaS products, as well as external contracting work for government and private organizations. Primarily worked with Javascript. Developed with technologies such as Node, Backbone, Marionette, jQuery, Elasticsearch, MongoDB, Angular, CouchDB, RethinkDB, Docker, and React.

Freelance Web Developer

April 2011 - November 2012 | *Kamloops, B.C.*

Independent Professional

Freelance web developer for a number of local businesses, organizations, and government entities. Many smaller projects focusing on simple web design for commercial businesses or community outreach, with a few larger projects consisting of full stack web applications made to handle large amounts of data and users. Websites were designed in raw HTML/CSS and web applications were built with MySQL, PHP, jQuery, and Backbone.

Education

Master of Science

Major in Computing Science

Research in Machine Learning and Computational Neuroscience

September 2016 - June 2019

University of Victoria, Victoria, British Columbia

Bachelor of Computing Science

Specialization in Software Engineering

Awarded the TRU Medal of Computing Science for highest graduating GPA

September 2012 - June 2016

Thompson Rivers University, Kamloops, British Columbia

Conferences and Publications

Using EGG to decode semantics during an artificial language learning task

Chris Foster, Chad C. Williams, Olave E. Krigolson, Alona Fyshe

Brain and Behavior, 2021

This research showed that EGG can be used to detect word meaning even during the process of language learning. Using electroencephalography and an artificial language that maps symbols to English words, we show that it is possible to use machine learning models to detect a newly formed semantic mapping as it is acquired. This work hints at new directions for studying brain function and the neural underpinnings of learning.

Words as a window: Using word embeddings to explore the learned representations of Convolutional Neural Networks

Dhanush Dharmaretnam, Chris Foster, Alona Fyshe

Neural Networks, Volume 137, 2021

This research shows different neural networks will build internal representations of knowledge which have similar patterns. We showed that the values in semantic models such as Skip-Gram have a correlation to the activations in convolutional neural networks like Inception-v3. Despite the fact that one is trained only on images and the other is trained only on text, they appear to learn similar representations of a word or image of the same semantic entity.

Improving Response Time Prediction for Stack Overflow Questions

Di Wu, Simon Johnson, Chris Foster, Erwin Li, Haytham Elmiligi, Musfiq Rahman

IEEE 10th Annual Information Technology, Electronics and Mobile Communication Conference

October 2019 - Vancouver, Canada

This research proposes a new machine learning model to predict the response times for questions posted on Stack Overflow. Our proposed problem formulation focuses on whether or not a question will receive an answer within a certain time frame, rather than predicting a specific interval of time within which an answer should be received. The paper also introduces a new set of features that help developers better characterize the properties of questions.

Decoding Music in the Human Brain using EEG Data

Chris Foster, Dhanush Dharmaretnam, Haoyan Xu, Alona Fyshe, George Tzanetakis

IEEE 20th International Workshop on Multimedia Signal Processing

August 2018 - Vancouver, Canada

This research project utilizes representational similarity analysis to measure the relationship between music features and the EEG data of subjects listening to the music. We show correlation between the mel spectrogram cepstral coefficients and tempogram features of the songs with the brain data of the subjects. We also use a simplified machine learning approach to achieve state-of-the-art classification of song titles from the EEG data on this dataset.

Decoding Word Semantics and Learning in EEG Data via an Artificial Language

Chris Foster, Chad C. Williams, Olave E. Krigolson, Alona Fyshe

Inaugural Conference on Cognitive Computational Neuroscience

September 2017 - New York, USA

This research project utilizes machine learning ridge regressors to predict word vector features based on input EEG data of a subject reading a word. The experiment is designed as a reinforcement learning paradigm, where the subject is viewing a series of symbols with a 1-to-1 mapping to an English word. As the subject learns the mappings we see word semantics begin to correlate to the brain's EEG readings as detected by a standard "2 vs. 2" test.

Response Time Predictions for Stack Overflow

Guojin Tang, Erwin Li, Chris Foster, Haytham El Miligi

12th Annual TRU Undergraduate Research and Innovation Conference

March 2017 - Kamloops, Canada

This research project surrounds the investigation of applications for machine learning algorithms on large web datasets. We explore the application of machine learning in the context of developer mailing list data and communities such as StackOverflow and Github. We improve on the work of a paper which predicts response times to Stack Overflow questions and explore ideas for improving the accuracy and applying to alternative datasets.

Tracking Cattle with Infrared Imaging Drones

Chris Foster, Matthew McInnes, John Church, Kevin O'Neil

10th Annual Irving K. Barber School Undergraduate Research Conference

April 2015 - Kelowna, Canada

This research project surrounds the development and investigation of a prototype quadcopter automated drone that can fly over ranching fields and identify cattle with a high-quality infrared camera and our custom computer vision algorithm. By utilizing the onboard GPS, we can generate a list of coordinates for the rancher. This is magnitudes cheaper for ranchers than the traditional method.

Scholarships and Awards

President's Research Scholarship

September 2016 - $4,000

University of Victoria

University of Victoria Fellowship

September 2016 - $13,500 (Declined)

University of Victoria

Outstanding Graduate Entrance Award

September 2016 - $5,000

University of Victoria

Canada Graduate Scholarship Master's Award

September 2016 - $17,500

Natural Sciences and Engineering Research Council of Canada

TRU Alumni Association Award

November 2015 - $1,200

Thompson Rivers University

British Columbia District Award

November 2012 - $500

Government of British Columbia

BC Interior Community Foundation Award

October 2012 - $300

BC Interior Community Foundation

British Columbia Provincial Award

September 2012 - $1,000

Government of British Columbia

Community Building

Victoria Machine Learning Meetup

February 2019 - August 2024

Lead Organizer

TRUCS Support Lab

September 2015 - December 2015

Support Lab Volunteer

TRUSU Computer Science Club

November 2014 - June 2016

Board Member - President

TRUSU Computer Science Club

March 2014 - November 2014

Board Member - Vice President

TRUSU Computer Science Club

November 2013 - March 2014

Board Member - Events Coordinator

Startup Weekend Kamloops

June 2013 - July 2015

Assistant Organizer

Projects

Javascript Bindings to a Spam Detection Service

June 2013 - Present

Lead maintainer of akismet-api, an npm package which provides convenient bindings to the akismet spam detection service. This package receives 70,000+ downloads per month and is relied upon by companies like Mozilla.

View on npm

Simple File Upload Service

February 2021 - Present

Lead maintainer of twostoryrobot/simple-file-upload, an docker image which provides an easy-to-deploy file upload service for kubernetes and other container environments. This image has been downloaded 100,000+ times.

View on Docker Hub

Community Talks and Presentations

Deep Learning for Self Driving Cars

Presented at the Thompson Rivers University Data Science Seminar

February 2021

Component Driven Development with Storybook

Presented at Startup Slam 2019

September 2019

Deep Learning for Self Driving Cars

Presented at the Victoria Machine Learning Meetup

July 2019

A Short Intro to Reinforcement Learning

Presented at Polyglot Unconference 2019

May 2019

Introduction to Reinforcement Learning

Presented at the Victoria Machine Learning Meetup

February 2019

Decoding Semantics During an Artificial Language Learning Task Using EEG

Presented at the University of Victoria Cognition and Brain Sciences Seminar

September 2017

The Fundamentals of Neural Networks
Presented at the Victoria Machine Learning Meetup
May 2017

Deep Learning for Self Driving Cars
Presented at the University of Victoria Advanced Software Engineering Research Seminar
April 2017

Video Super Resolution with Deep Learning
Presented to the University of Victoria Computing Science department
April 2017

SwoopVR: A Web-based Mesh Viewer
Presented to Thompson Rivers University faculty
April 2016

Mining Stack Overflow
Presented to Thompson Rivers University faculty
April 2016

Introduction to Functional Programming
Presented to the TRUSU Computer Science Club
February 2016

Introduction to Automated Testing
Presented to the TRUSU Computer Science Club
November 2015

Tracking Cattle with Infrared Imaging
Presented to Thompson Rivers University faculty
May 2015

Introduction to Python
Presented to the TRUSU Computer Science Club
April 2015

Creating Zero-Knowledge SaaS Applications
Presented at the Okanagan Developers Group Meetup
October 2014

Introduction to Git
Presented to the TRUSU Computer Science Club
October 2014

Introduction to Cryptography
Presented to the TRUSU Computer Science Club
September 2014

Certifications and Courses

How to Manage a Remote Team
October 2020
Coursera / GitLab

AWS Certified Solutions Architect Associate
September 2017
Amazon Web Services

Standard First Aid & CPR
January 2015
Canadian Red Cross

MongoDB Developer Course Certification
December 2012
MongoDB Inc.

Machine Learning Course Certification
December 2011
Coursera / Stanford University

Research Interests

- Artificial intelligence, machine learning
- Privacy, applied cryptography, secure communications, and zero-knowledge services
- Software development, open source software, web technologies
- Security in mobile applications, web applications, and networks
- Practical applications of new types of software for fields other than computing science

Personal Achievements

- Represented Canada at the 2025 Ironman World Championship in Nice, FR
- Placed 1st overall at the 2025 CJFC TV Boogie the Bridge 10km Run
- Represented Canada at the 2024 Ironman 70.3 World Championship in Taupo, NZ
- Placed 4th in the M:30-34 division at the 2024 Ironman Canada Triathlon
- Placed 2nd overall at the 2024 Yuma Territorial Marathon
- Placed 2nd in the M:20-29 division at the 2022 Across the Lake Skaha Swim
- Placed 1st in the M:18-24 division at the 2018 Oliver Half Iron Triathlon
- Placed 1st in the M:20-24 division at the 2016 Oliver Half Iron Triathlon
- Placed 2nd in the M:20-29 division at the 2016 Kamloops Sprint Sprint Triathlon
- Placed 3rd in the M:20-29 division at the 2015 Pavillion Sprint Triathlon
- Placed 1st in the M:20-29 division at the 2015 Kamloops Sprint Sprint Triathlon
- Placed 3rd in the M:20-29 division at the 2014 Pavillion Sprint Triathlon
- Placed 1st in the M:16-19 division at the 2013 Pavillion Sprint Triathlon

Languages

English
Native Speaker

Esperanto
Elementary Proficiency

Spanish
Elementary Proficiency